

THA PHRA AGROMET, Thailand

WMO: 483840

Lat: 16.335N

Lon: 102.827E

Elev: 545

StdP: 14.41

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	56.7	59.9	48.6	51.7	64.4	51.9	58.4	65.2	9.4	77.0	8.3	75.4	0.7	N/A	0.298

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	18.2	101.2	77.3	98.9	77.9	96.7	78.1	82.5	91.6	81.8	90.6	81.3	89.9	3.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.4	160.8	86.2	79.8	157.9	85.5	79.2	154.7	84.8	46.8	91.6	46.1	91.1	45.6	90.3	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.2	5.9	DB	51.0	103.1	3.3	2.9	48.7	105.2	46.7	106.9	44.8	108.5	42.4	110.6	(4)
(5)			WB	49.5	84.2	3.6	1.0	46.9	84.9	44.8	85.5	42.8	86.1	40.2	86.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.8	74.8	79.0	83.8	86.9	85.8	85.2	83.7	83.0	82.7	81.9	79.6	75.3
	DBStd	5.33	4.83	5.27	5.28	4.49	3.20	2.30	2.83	2.18	2.05	2.35	3.49	4.83
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	4	3	0	0	0	0	0	0	0	0	0	0	1
	CDD50	11605	768	811	1048	1107	1109	1056	1043	1024	981	988	887	783
	CDD65	6134	306	391	583	657	644	606	578	559	531	523	437	319
	CDH74	66568	2519	3916	7031	8492	7795	7382	6523	5921	5406	5047	3903	2633
	CDH80	29397	939	1868	3733	4744	3764	3337	2578	2076	1859	1899	1590	1010
(14) Wind	WSAvg	1.2	1.2	1.1	1.3	1.3	1.1	1.4	1.5	1.3	0.8	0.9	1.2	1.3
(15) Precipitation	PrecAvg	48.7	0.3	0.7	1.7	3.7	6.4	5.9	7.0	8.7	9.6	3.9	0.7	0.2
	PrecMax	68.5	2.5	2.7	4.9	11.3	11.6	12.5	12.9	15.9	17.8	7.6	4.2	1.5
	PrecMin	35.6	0.0	0.0	0.0	0.7	2.0	2.1	1.1	3.3	3.0	0.3	0.0	0.0
	PrecStd	8.0	0.6	0.8	1.5	2.6	2.6	2.3	2.6	2.9	3.5	2.2	1.1	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.1	99.1	102.9	105.0	101.2	97.4	98.5	94.2	93.6	94.2	94.7	94.1
		MCWB	73.7	75.2	74.3	78.0	79.8	80.0	79.8	79.9	81.1	79.5	78.4	75.0
	2%	DB	90.6	96.2	100.4	102.3	98.3	95.2	94.5	92.0	91.9	92.1	92.2	91.4
		MCWB	73.6	74.9	75.5	78.2	79.9	80.1	79.8	80.1	80.7	79.0	77.6	74.9
	5%	DB	88.2	93.5	97.9	99.5	95.7	93.2	91.9	90.2	90.2	90.5	90.4	88.7
		MCWB	73.0	74.5	76.2	78.6	80.3	80.1	79.7	80.0	80.4	78.8	76.9	73.8
	10%	DB	85.6	90.5	95.1	96.7	93.2	91.4	89.5	88.5	88.5	88.9	88.3	85.9
		MCWB	72.2	74.4	76.5	78.7	80.5	80.0	79.2	79.7	80.1	78.4	76.2	72.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.8	79.3	80.8	82.4	83.6	83.1	81.8	82.4	82.7	81.8	80.6	77.7
		MCDB	87.1	92.1	94.8	95.7	93.3	91.8	90.8	89.1	90.2	90.0	90.2	89.6
	2%	WB	76.3	77.7	79.5	81.4	82.4	82.0	80.9	81.5	81.8	80.8	79.2	76.1
		MCDB	85.4	90.4	93.0	94.4	92.6	90.4	89.4	88.2	88.8	88.7	89.0	87.8
	5%	WB	74.9	76.6	78.5	80.6	81.6	81.3	80.3	80.8	81.1	79.9	78.4	74.9
		MCDB	83.4	88.6	91.4	93.2	91.2	89.5	88.3	87.6	87.8	87.3	87.5	85.8
	10%	WB	73.6	75.5	77.7	79.7	80.9	80.6	79.8	80.1	80.5	79.2	77.4	73.5
		MCDB	82.4	86.9	89.8	91.8	89.9	88.6	87.3	86.4	86.8	85.8	84.7	83.1
(35) Mean Daily Temperature Range	5% DB	MDBR	21.9	22.7	20.6	18.2	15.0	13.0	11.7	11.4	12.0	14.3	17.9	21.3
		MCDBR	24.0	25.1	23.9	20.6	17.9	15.4	15.2	13.8	14.5	16.4	19.5	22.9
		MCWBR	8.5	7.2	5.9	5.1	4.7	4.3	4.5	4.8	5.1	5.7	6.3	7.9
	5% WB	MCDBR	19.7	20.9	20.1	18.5	16.1	14.3	13.3	12.6	13.1	14.5	17.2	20.8
		MCWBR	8.0	6.9	6.3	5.6	5.5	4.9	4.6	5.1	5.3	5.7	5.9	7.7
(40) Clear-Sky Solar Irradiance	taub		0.479	0.589	0.656	0.615	0.510	0.461	0.450	0.459	0.503	0.539	0.477	0.471
	taud		2.075	1.807	1.694	1.830	2.150	2.307	2.339	2.301	2.164	2.027	2.170	2.120
	Ebn at Noon		255	233	222	232	254	264	267	267	257	243	254	253
	Edh at Noon		50	69	79	69	49	42	41	43	49	54	45	47
(44) All-Sky Solar Radiation	RadAvg		1547	1646	1680	1813	1750	1643	1533	1496	1466	1563	1555	1502
	RadStd		93	81	83	74	92	84	106	90	97	112	58	39

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)	+0.54	N/A	N/A	N/A	+0.31	+0.25	N/A	N/A	+216	+220

Nomenclature: See separate page